Project Report

1. Display your output of `plot_Kmeans()`. Does your plot match your expectations?

   the output is a file in the directory. It is very much as expecteded the numbers look handwritten and i can tell which is which.

2. In this assignment, you implemented k-means using a Euclidean distance metric. Describe other distance metrics that can be used and how they cluster inputs.

   - Manhattan distance: Sums the differences across dimensions. It is less sensitive to large outliers than Euclidean distance and can produce more secure clusters when data contains noise or extreme values.
   - Cosine distance: Measures the angle between two vectors rather than their magnitude. This is useful when the overall brightness of an image varies between examples, since two images of the same digit drawn with different pen pressing would still be considered similar.
   - Chebyshev distance: Takes the maximum absolute difference across all dimensions. This emphasizes the single most different pixel, making clusters more sensitive to local extremes.

3. What would you expect the cluster centers (centroids) to look like if we use $K < 10$? $K > 10$?

   $K < 10$: With fewer clusters than digits, some centroids would be forced to represent more than one digit. These centroids would look like blurry blends of two or more similar digits — for example, a single centroid might try to capture both 3s and 8s, producing a smeared image that clearly resembles neither. The accuracy would drop since the classifier cannot distinguish all 10 classes.

   $K > 10$**:** With more clusters than digits, some digits would be split across multiple centroids. For instance, digit 1s written with a serif might form one cluster while slanted 1s form another. The centroids would look crisper and more specific, but the manual assignment step becomes harder since multiple centroids would need to map to the same digit. This could actually improve accuracy if the extra clusters capture meaningful style variation, but risks overfitting.